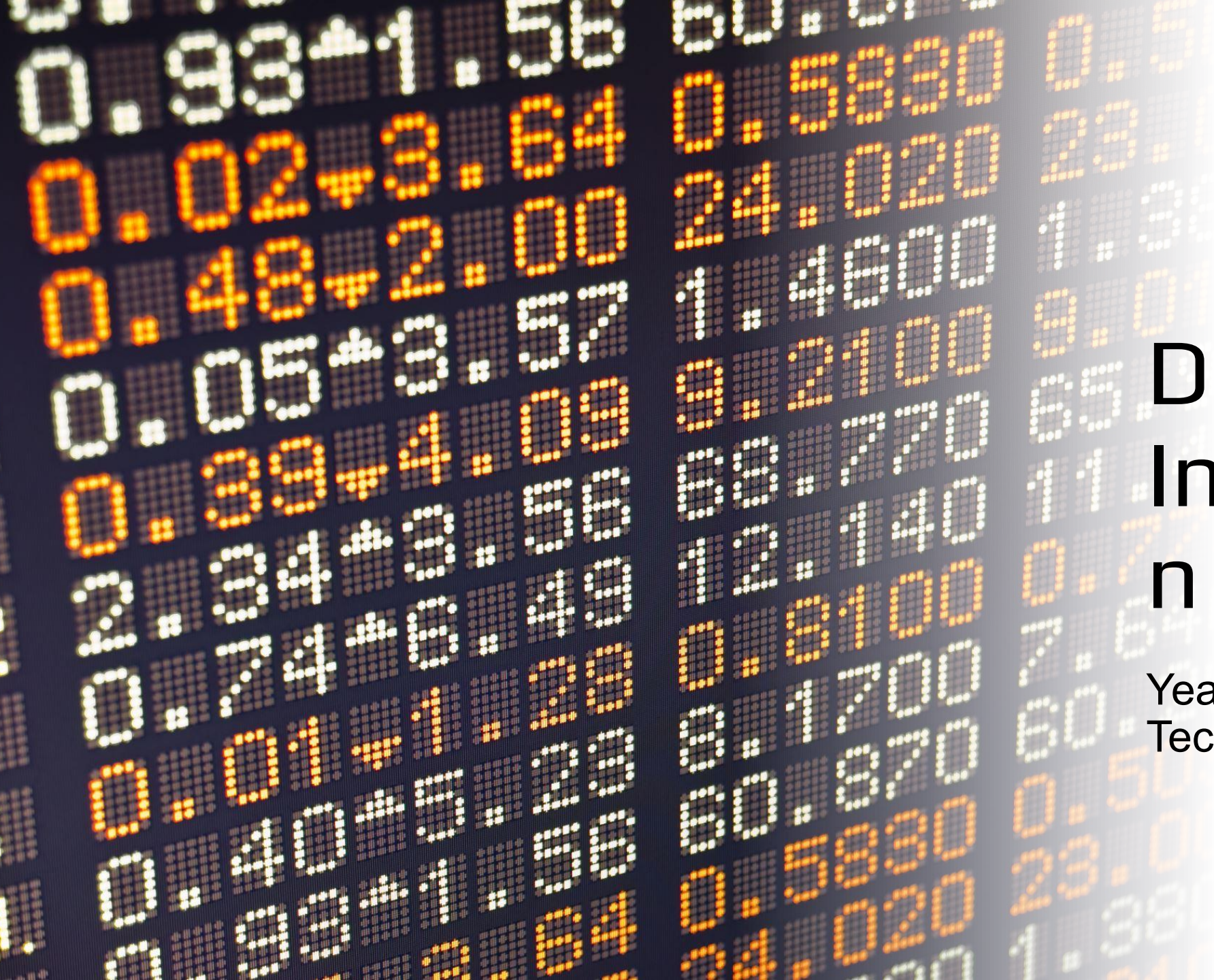


Data and Information

Year 9 Computing Technology

Learning Intention



Differentiate between data and information



Describe the purpose of analysing data



Describe inputs, storage, transmission, processes and outputs in data analysis



Represent and store data to facilitate computation, including selecting appropriate data types, understanding data type limitations and structuring data systematically

Success Criteria



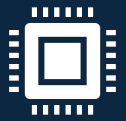
I can explain the difference between data and information, using examples.



I can describe the purpose of analysing data.



I can memorise and describe inputs, storage, transmission, processes and outputs in data analysis



I can describe a variety of different data types commonly used in information systems.

Data and Information

Data and information are everywhere in daily life

Humans collect, process, store and use information constantly

Our brains have limits, so we rely on **digital technology**

Computers help collect, process, store and transfer data quickly

Unlike humans, computers must be told **exactly what to do**

Data vs Information

Data: Data is raw, unorganized, and meaningless facts, symbols, or figures

Information: Information is processed, structured, and contextualized data that has meaning and is useful for decision-making

Examples:

A list of numbers = data

Being told they are phone numbers = information

Data becomes information when **people interpret it**

Data in Computer Systems



Computers require data in **digital form**



Information becomes data when entered into a computer



Data becomes information again when displayed and understood



Computers only see patterns of **binary data (0s and 1s)**



Meaning is added by the **user**, not the computer

Digital Data and Binary



All digital data is made of **bits** (0 or 1)



Binary allows computers to process data quickly and accurately



Large amounts of data can be stored and processed rapidly



A group of 8 bits is called a **byte**



One byte can represent **256 different values**

Analog vs Digital Data

Analog data: changes smoothly (sound volume, speed, light)

Digital data: fixed, counted values

Computers work only with **digital data**

Analog data must be **digitised** before processing

Example: microphone sound converted by a sound card

Data Coding



Computers use **binary coding** to represent data



The same binary value can mean different things



Meaning depends on how the data is used



Binary data can represent:

- Text
- Colours
- Sounds
- Instructions



Context determines interpretation

Data Types and File Formats

- A **file format** defines how data is organised
- Same bits can represent different data types
- File formats tell software how to interpret the data
- Common data types include:
  - Text
  - Graphics
  - Audio
  - Animation
  - Video





Text and Hypertext



Text: characters, symbols and spaces



Stored and processed digitally



Hypertext: text that links to other documents or locations



Used widely on the internet (web pages, documents)

Graphics

Graphics are images used to communicate information

Bitmap: made of pixels

Vector: made using mathematical shapes

Colour depth affects image quality and file size

More colours = larger file size

Audio



Audio is **digitised sound**



Represents sound waves using binary values



Used for speech, music and sound effects



Audio files can be large



Compression reduces file size

Animation

- Animation is a sequence of images shown quickly
- Creates the illusion of movement
- Often computer-generated
- Techniques include morphing and warping
- Used in games, films and simulations

Video



Video is a series of images of the real world



Often includes audio



File sizes can be very large



Streaming allows playback while downloading



Common in media, education and communication

Quantitative and Qualitative Data



Quantitative data is data that can be counted or measured in numerical values



Qualitative data is descriptive data that is not expressed numerically.



Both types are valuable depending on purpose



AI can help analyse qualitative data

Data Analysis

Summarise all



Input:
Raw data is entered into a computer system using input devices such as keyboards, sensors, scanners or microphones.



Process:
The computer manipulates the data using software instructions to sort, calculate, analyse or transform it.



Storage:
Data and information are saved in storage devices or databases so they can be accessed and used later.



Transmission:
Data is transferred between systems or devices using networks such as the internet or local networks.



Output:
Processed data is presented as information that people can understand, such as text, images, sound or video.

Common Uses of Data



Data is used to:

- Share information
- Control systems (traffic lights, machines)
- Analyse trends
- Make predictions



AI systems rely on large datasets



More data generally leads to more accurate results